(3) If $f(x) \leq 0$ for all $x \in [a,b]$, then the Riemann integral $\int_a^b f(x)dx$ is equal to the negative of the geometric area of the region bounded by the graph of $y = f(x)$, the x-axis, the lines $x = a$ and $x = b$. See Fig. 9.4. In this case, the geometric area of the region bounded by the graph of $y = f(x)$, the x-axis, the lines $x = a$ and $x = b$ is given by $\left|\int_a^b f(x)dx\right|$.

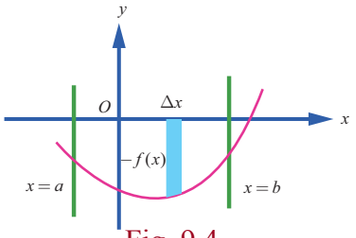


Fig. 9.4

(4) If $f(x)$ takes positive as well as negative values on $[a,b]$, then the interval $[a,b]$ can be divided into subintervals $[a,c_1]$, $[c_1,c_2], \cdots, [c_k,b]$ such that $f(x)$ has the same sign throughout each of subintervals. So, the Riemann integral $\int_a^b f(x)dx$ is given by

$$\int_a^b f(x)dx = \int_a^{c_1} f(x)dx + \int_{c_1}^{c_2} f(x)dx + \cdots + \int_{c_k}^b f(x)dx .$$

In this case, the geometric area of the region bounded by the graph of $y = f(x)$, the x-axis, the lines $x = a$ and $x = b$ is given by

$$\left|\int_a^{c_1} f(x)dx\right| + \left|\int_{c_1}^{c_2} f(x)dx\right| + \cdots + \left|\int_{c_k}^b f(x)dx\right| .$$

For instance, consider the following graph of a function $f(x), x \in [a,b]$. See Fig. 9.5. Here, A_1, A_2 and, A_3 denote geometric areas of the individual parts. Then, the definite integral $\int_a^b f(x)dx$ is given by

$$\begin{aligned}\int_a^b f(x)dx &= \int_a^{c_1} f(x)dx + \int_{c_1}^{c_2} f(x)dx + \int_{c_2}^b f(x)dx \\ &= A_1 - A_2 + A_3 .\end{aligned}$$

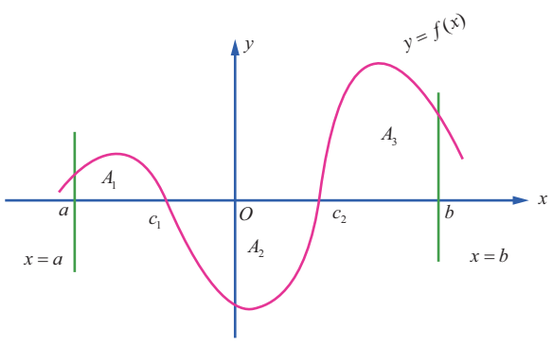


Fig. 9.5

The geometric area of the region bounded by the graph of $y = f(x)$, the $x-$axis, the lines $x = a$ and $x = b$ is given by $A_1 + A_2 + A_3$. In view of the above discussion, it is clear that a Riemann integral need not represent geometrical area.

Note

Even if we do not mention explicitly, it is always understood that the areas are measured in square units and volumes are measured in cubic units.

Example 9.1

Estimate the value of $\int_0^{0.5} x^2 dx$ using the Riemann sums corresponding to 5 subintervals of equal width and applying (i) left-end rule (ii) right-end rule (iii) the mid-point rule.

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